

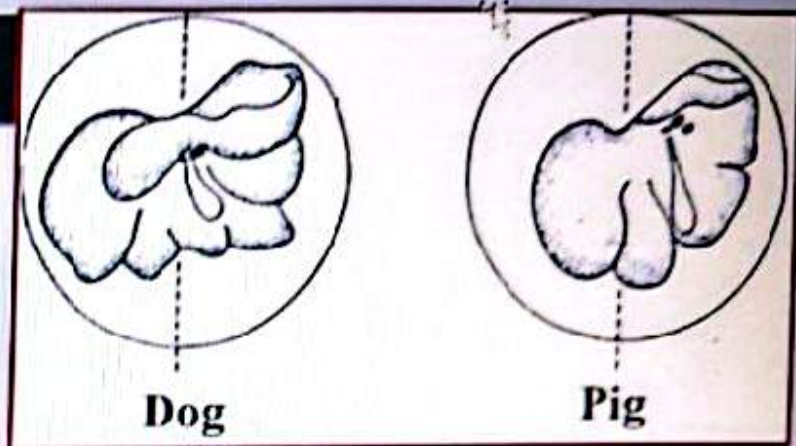
Liver

Definition:

- ✓ Largest gland in body
- ✓ Belong to digestive system.

Position:

- ✓ Situated in intra-thoracic part of abdominal cavity



Non-ruminants

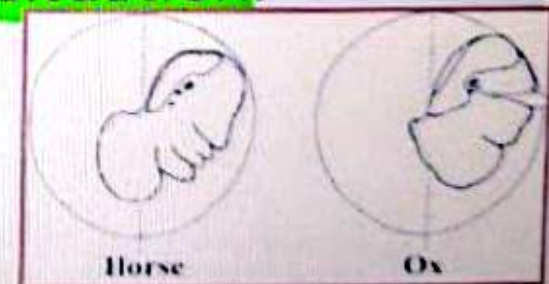
dog, pig, and horse

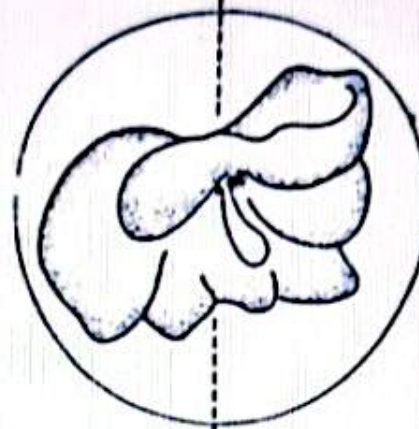
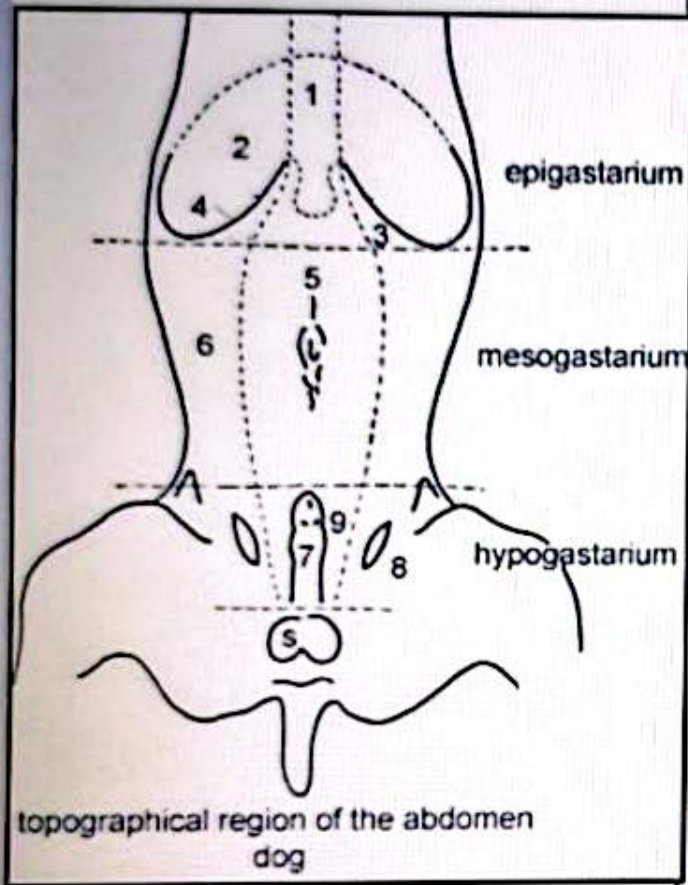
- In epigastric region
- In transverse plane
- behind diaphragm

ruminants

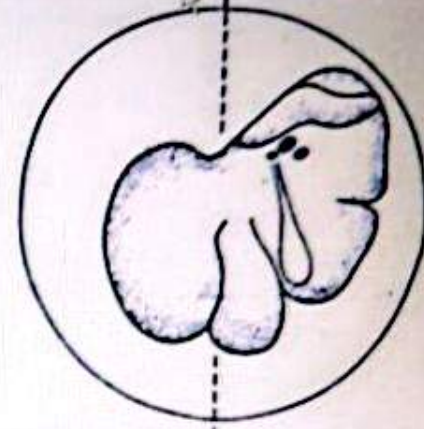
Ox, sheep, goat, and camel

- In right hypochondriac region
- with oblique longitudinal axis (craniocaudally).
- In a sagittal situation.

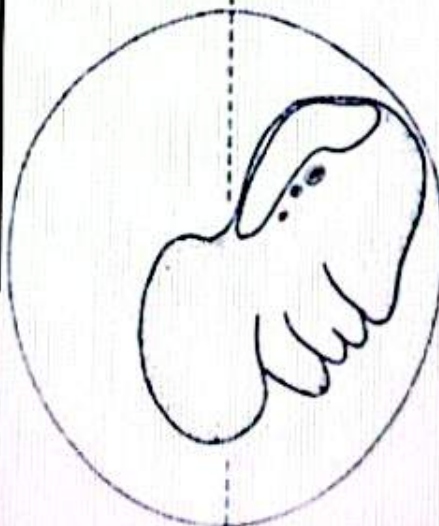




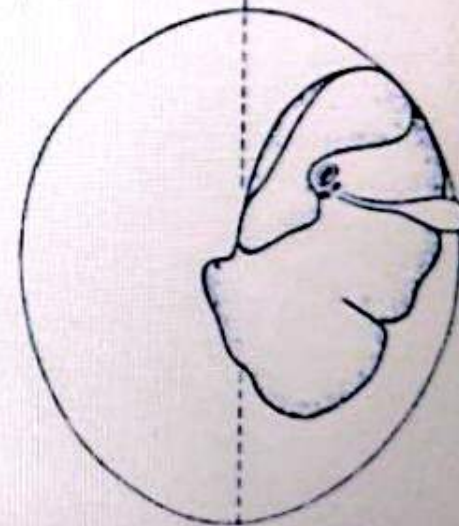
Dog



Transverse plane



Horse



Ox

Sagittal plane

Functions:

1. Secretion of bile for fat digestion.
2. Metabolic process (fat, carbohydrate, protein).
3. Storage fat, glycogen and vitamins.
4. Blood formation (RBCs in fetal life, blood protein)
5. Secretion of heparin.
6. Detoxication.

Color: In general, liver is reddish brown in color.

Depend on:

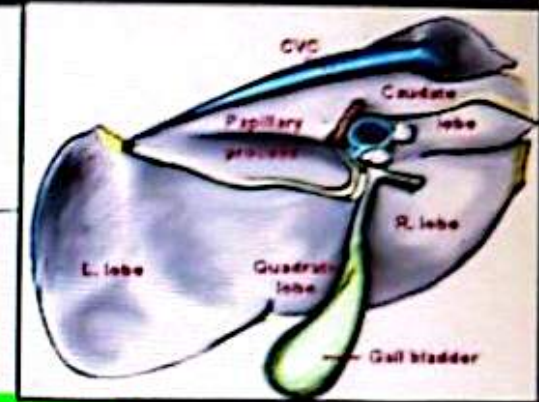
- A. *Amount of blood it contains*
- B. *Species*
- C. *Age*
- D. *Nutritional state of animal.*

Size and weight:

Depend on:

- A. *Species*
- B. *Age*
- C. *Condition of animal .*

Gross anatomy of liver



■ Has:

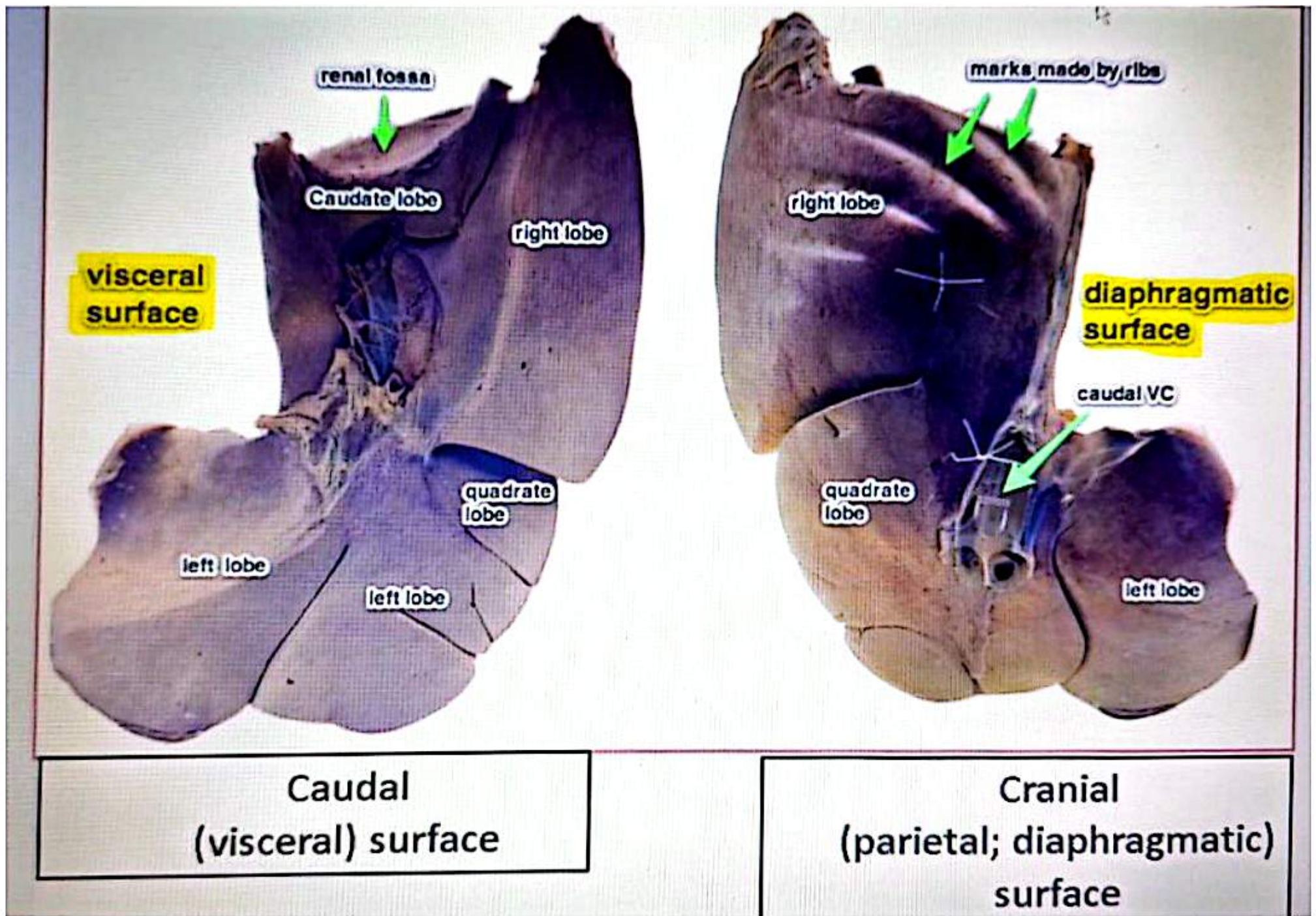
✓ Two surfaces

1) Cranial (parietal; diaphragmatic) surface:

- ❖ Convex
- ❖ Adapted to abdominal surface of diaphragm.

2) Caudal (visceral) surface

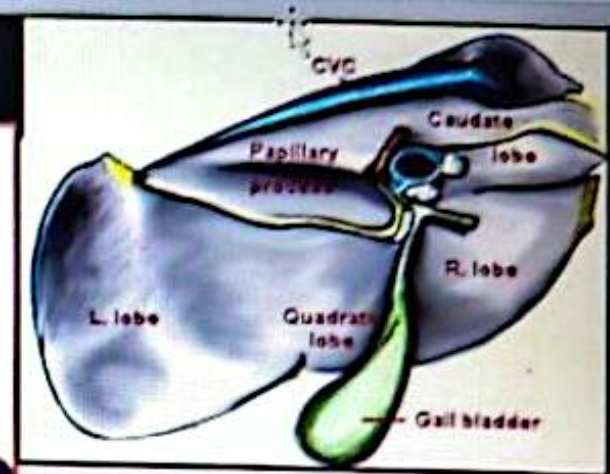
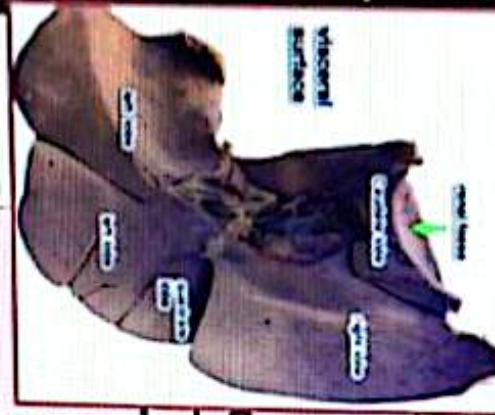
- ❖ Concave
- ❖ Show number of **impressions** relevant to adjacent viscera
- ❖ At its middle: **portal area** is found in which **portal V., hepatic A. & nerve enter liver** and **lymph vessel and bile duct leave it.**



Two borders (dorsal & ventral)

1) Dorsal border:

- ❖ Rounded
- ❖ Formed by union of cranial & caudal surfaces dorsally
- ❖ Shows:
 - ❑ Esophageal impression on left side
 - ❑ Renal impression on right side
 - ❑ Caudal vena cava impression between them.



2) Ventral or acute border:

- ❖ Thin and sharp
- ❖ Shows:
 - ❑ Umbilical fissure for round ligament on left
 - ❑ Gall bladder fossa appears on right.

oesophageal groove

caudal vena cava groove

papillary lobe

caudate lobe

Porta hepatis

left
medial
lobe

quadrate lobe

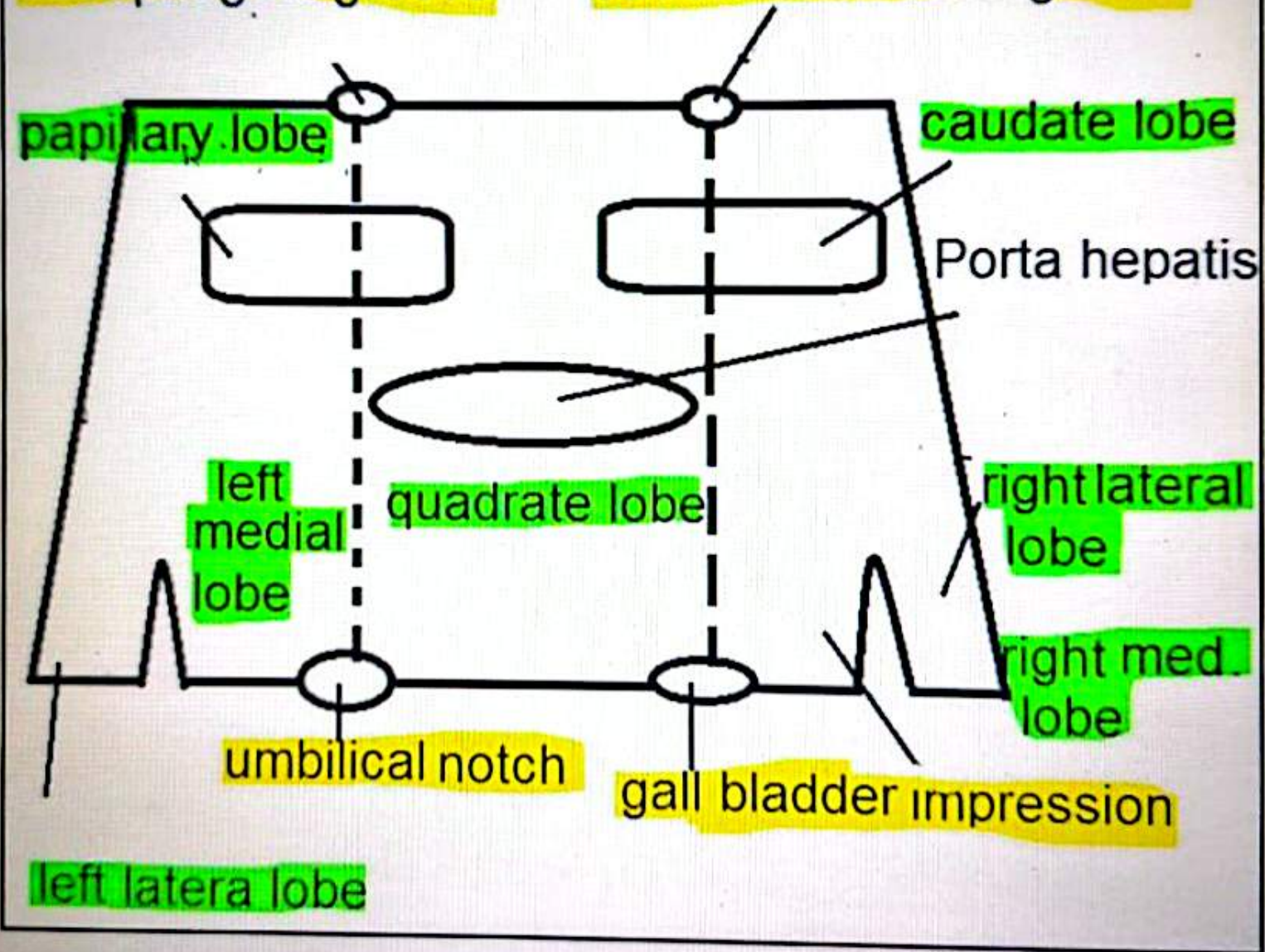
right lateral
lobe

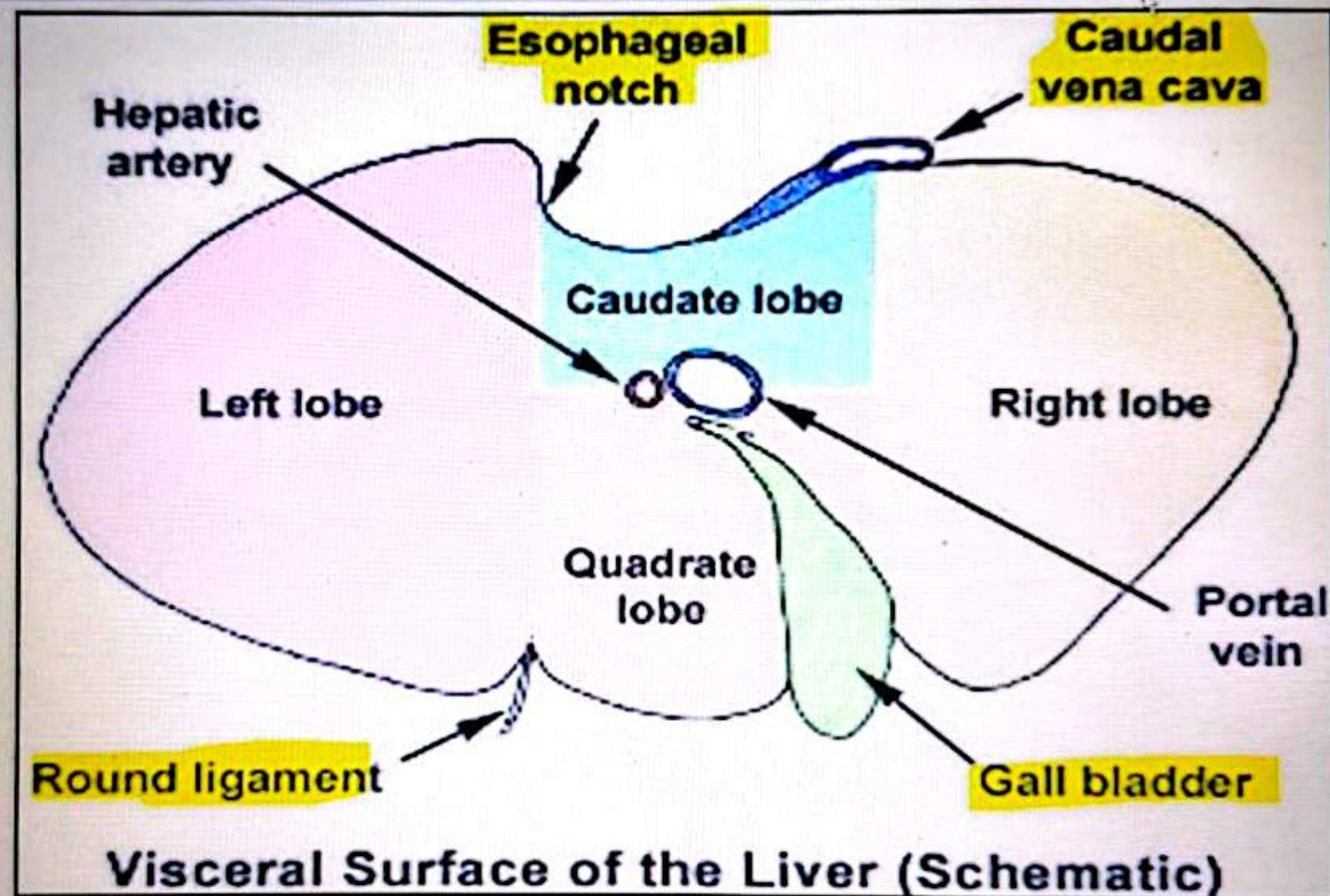
right med.
lobe

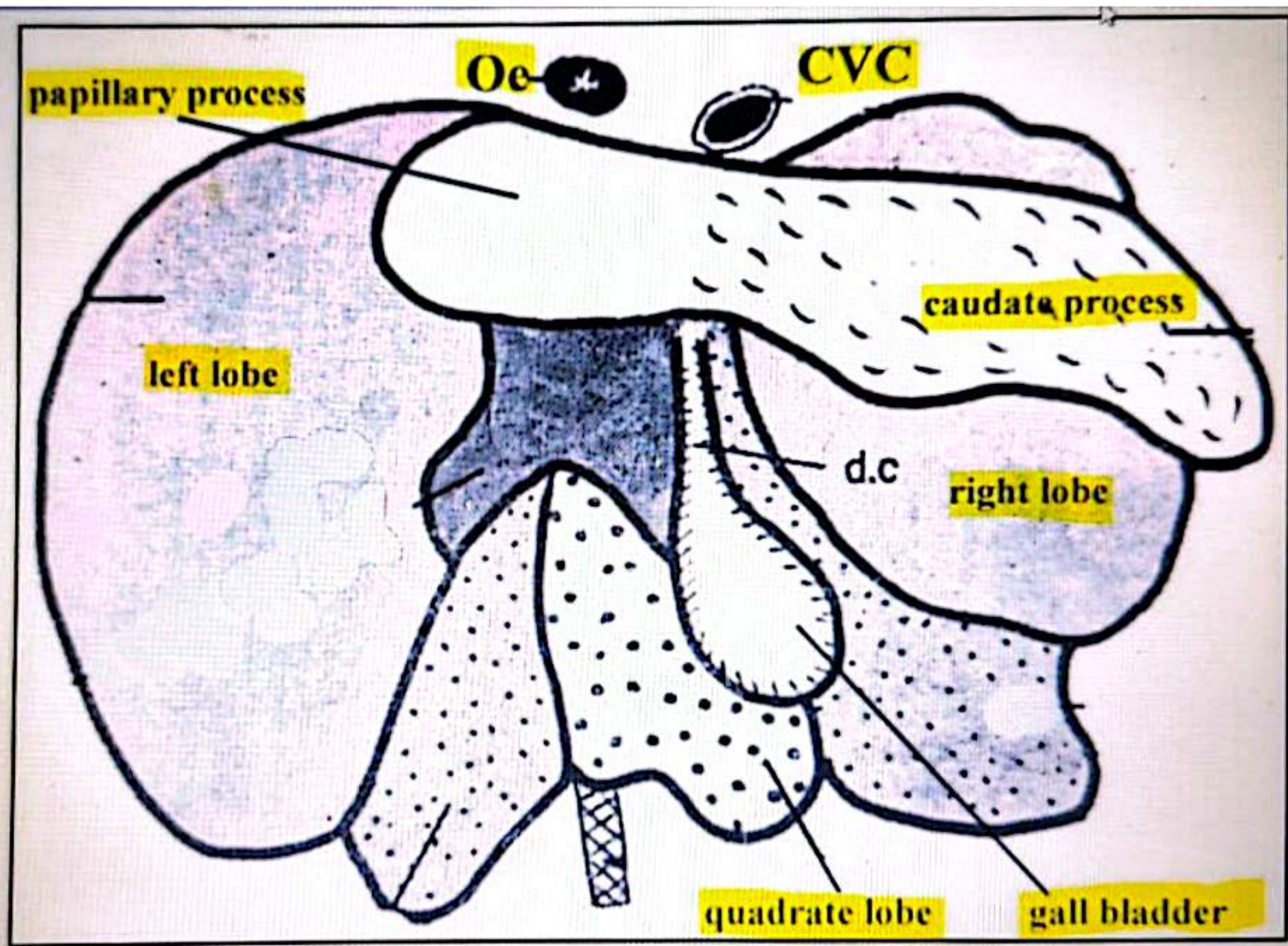
umbilical notch

gall bladder impression

left lateral lobe







Fixation of the liver

I. Non-peritoneal fixation:

1. Connection with large bl. Vs+ cd V.C. +portal vein
2. Pressure of the surrounding organs.

Fixation of the liver

II. Peritoneal fixation:

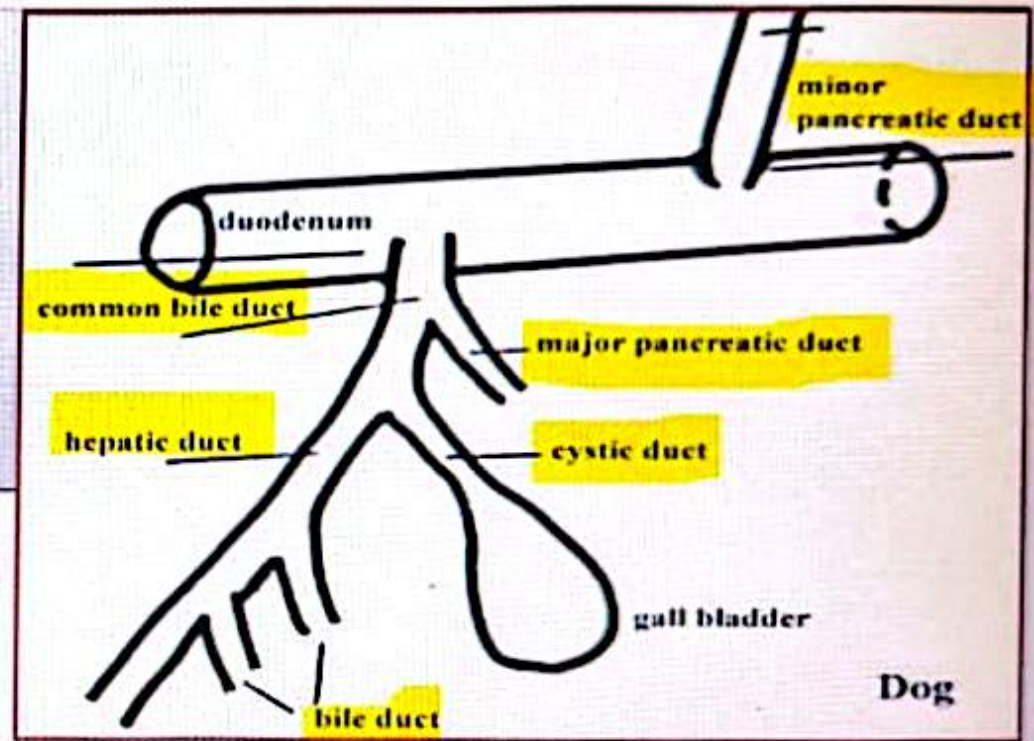
1. **Left triangular lig.:** attached to adjacent part of **diaphragm**.
2. **Right triangular lig.:** attached to adjacent part of **diaphragm**.
3. **Coronary lig.:** attach parietal surface of liver to **diaphragm**.
4. **Falciform lig.:** peritoneal fold extend from umbilical fissure of liver to the **sternal part of diaphragm**.
5. **Round lig.:** on free border of **falciform ligament**, & represents the **obliterated umbilical vein of the fetus**.
6. **Hepato-renal:** from **caudate lobe & right lobes** to **right kidney**
7. **Lesser omentum:** two ligaments:
 - A. **Hepatogastric:** from liver at **porta hepatic** to stomach
 - B. **Hepatoduodenal:** from liver at **porta hepatic** to duodenum

Ligaments	Dog	Pig	Rum.	Horse	Camel
R. & L. triangular	Right: well-developed -Left: small	Both absent	Both well-developed		
Coronary	Well-developed				developed
Hepatorenal	Present	Absent	Present		
Falciform	Small	Absent in adult	Absent in adult	Present	Present
Round	Present	In young	In young	Present	Present
Lesser omentum	From liver to stomach & duodenum		From liver to omasum, abomasum & duodenum	As dog	As rum.

Excretory apparatus of the liver

■ Consists of:

1. Common hepatic duct
2. Gall bladder
3. Cystic duct
4. Common bile duct.



Excretory apparatus of the liver

Bile leaves liver through lobar ducts (small bile ducts)



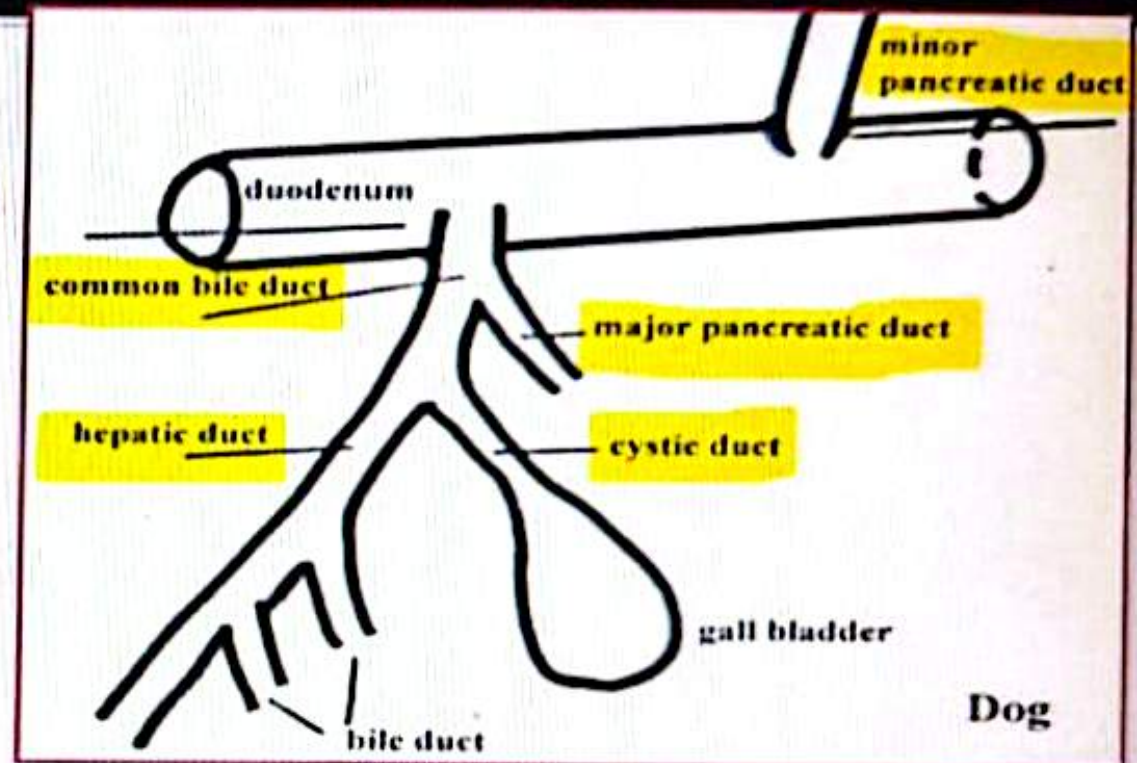
That united forming common hepatic duct



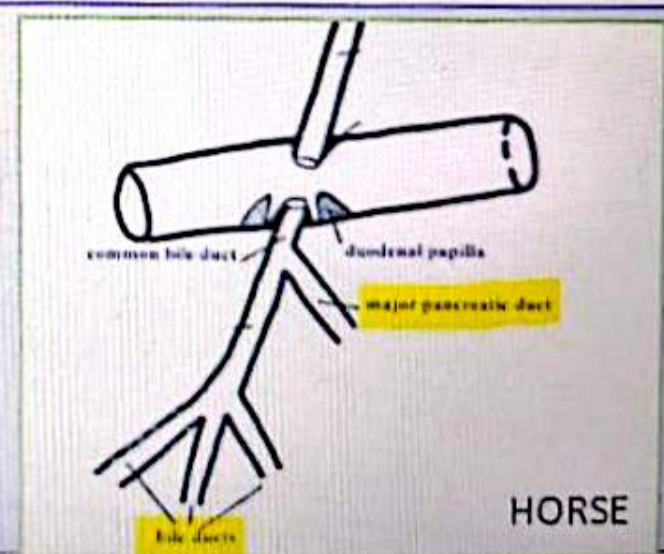
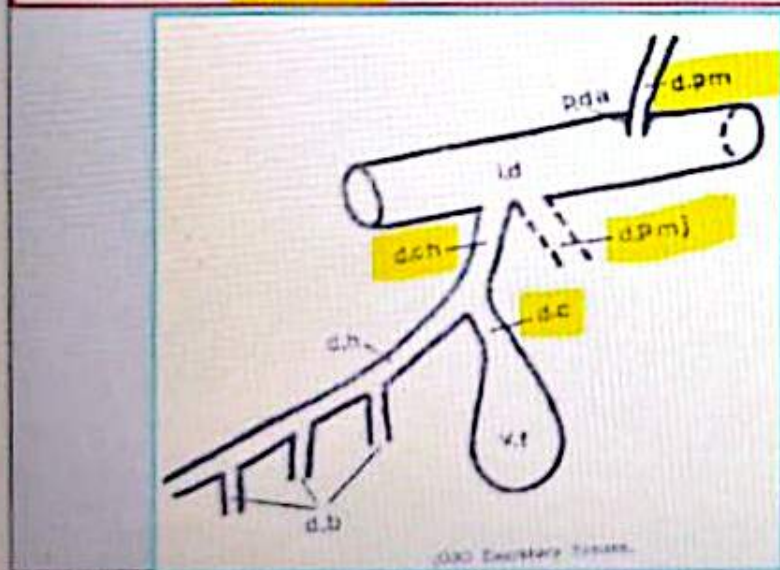
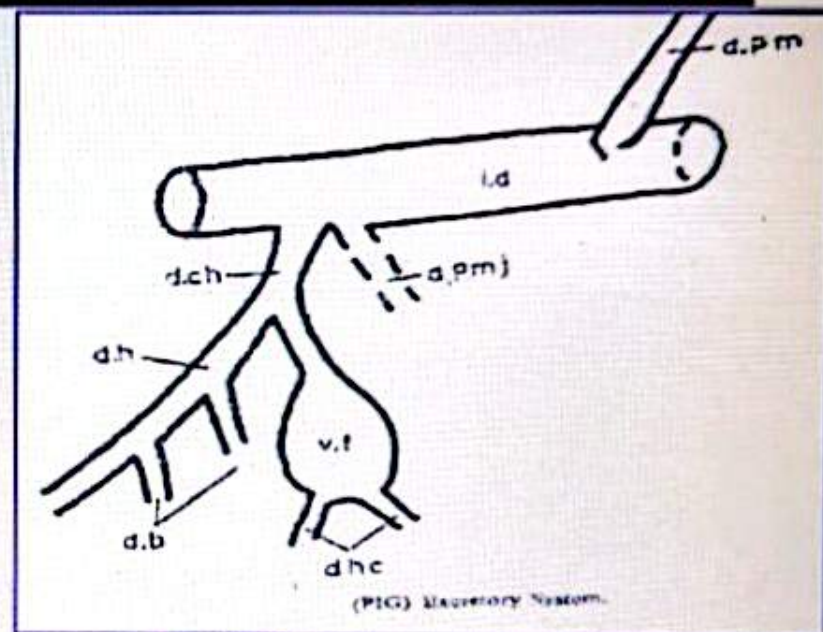
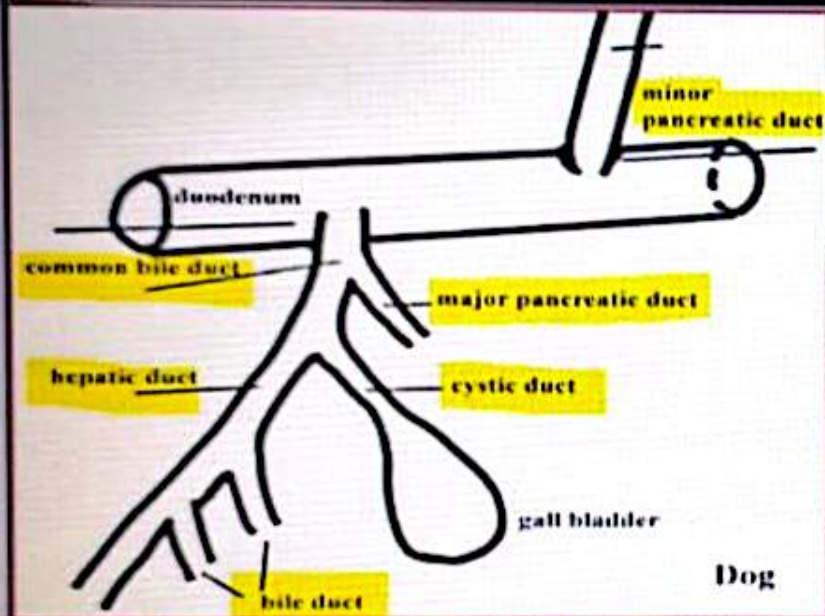
That united with cystic duct (run in hepato-duodenal ligament)



To open together with major pancreatic duct (horse, camel, sheep, and dog) on duodenal papillae.



Excretory apparatus of liver



Gall bladder

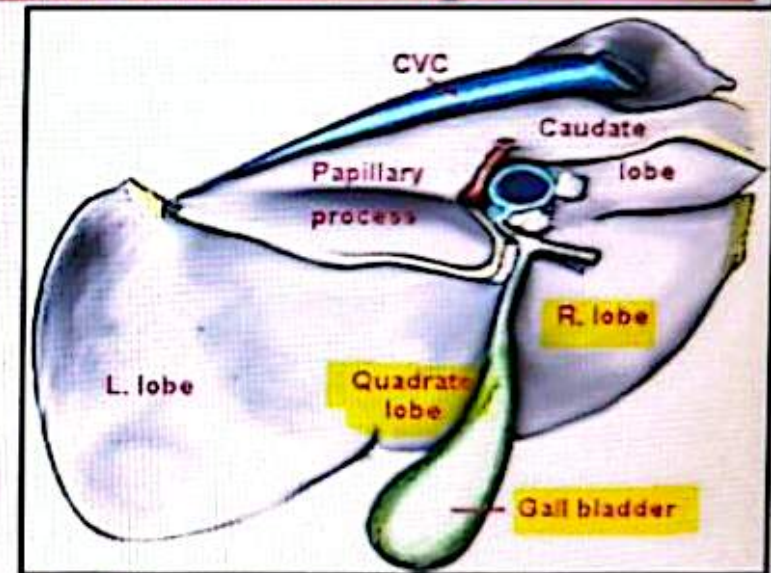
- Pear shaped sac
- Act as a reservoir for excretory apparatus
- Located on fossa between quadrate lobe & right medial lobe of liver

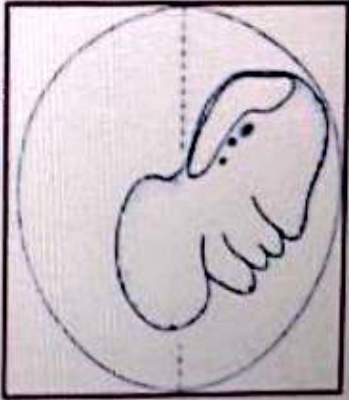
- Consists of:

1. Fundus
2. Body
3. Neck which continue by cystic duct

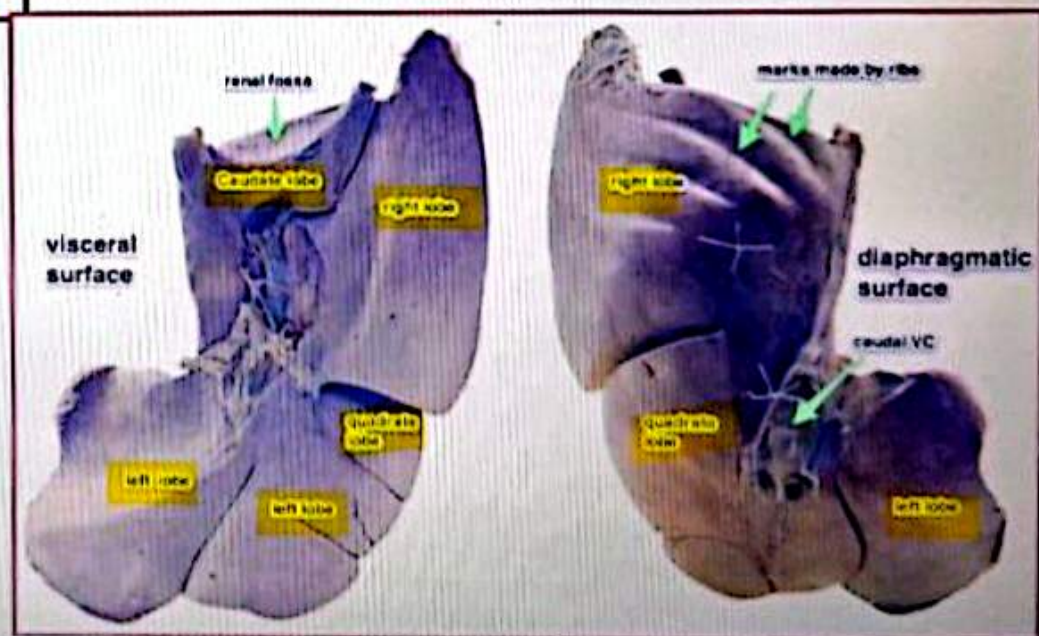
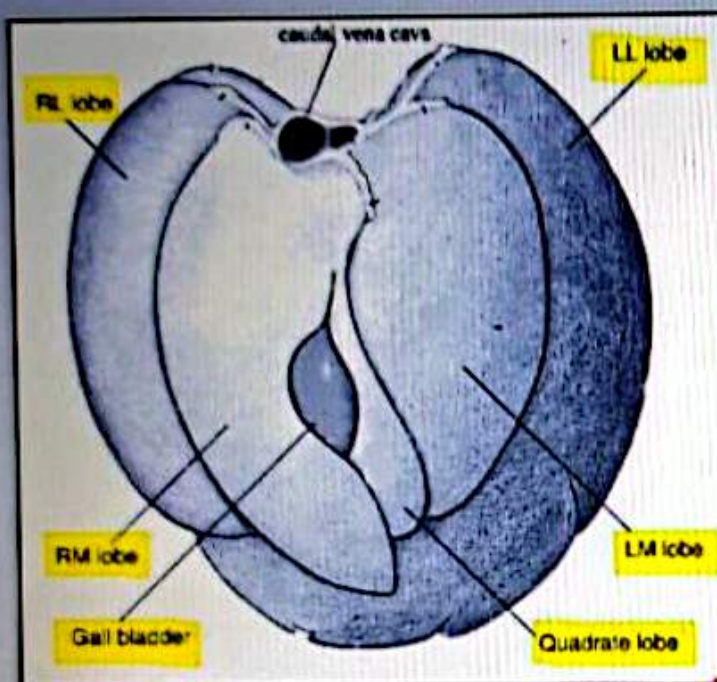
- Abscent in equine + camel

➤ In which CHD dilated and considered as CBD



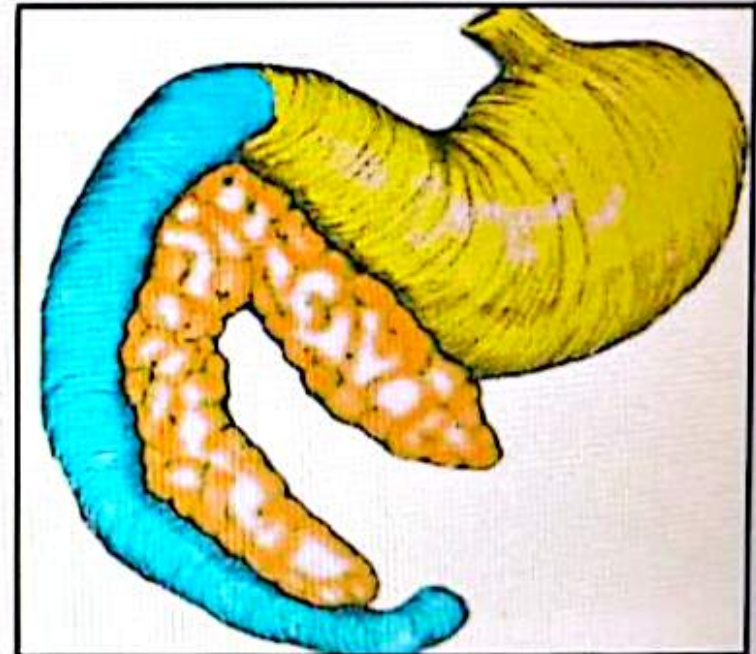
	DOG	PIG	HORSE
		Centrally	Obliquely 3/5 to right, 2/5 to left
Position		Intrathoracic part of abdominal cavity ➤ Epigastric region	
Lobulation	• Well-developed by deep interlobar fissure • 7 lobe	• Deep interlobar fissure • 6-lobe • (divided—right & divided—left+ quadrate+ quadate)	• Poor lobation • Called undivided liver • Quadrate lobe fissured at ventral border • Right lobe undivided • Left lobe divided
Renal impression	• Present	• absent	• Present
Gall bladder	• Bet. quadrate & right medial lobe • Not reach ventral border of liver • Visible on parietal surface of liver		• Abscent 

	OX	CAMEL
Position	- Intrathoracic part of abdominal cavity - Right hypochondriac region	
Lobulation	5 lobes - Undivided left - undivided right - Quadrate - Caudate lobe divided into: - Large caudate process - Small papillary process	- Lobes not clear - Triangular liver - Apex cranially <u>Liver hard in texture due to well-developed interlobular C.T.</u> - Visceral surface fissured as cutted by sharp knife
Renal impression	Present	Present
Gall bladder	pear shape	abscent



Pancrease

- **Def.:**
 - ✓ Encapsulated and lobulated gland
 - ✓ Has exocrine & endocrine function
- **Position:** in epigastric region
- **Consists of:**
 - ✓ **Left lobe;** lie dorsal to stomach
 - ✓ **Right lobe:** related to duodenum
 - ✓ **Middle lobe:** related to the beginning of duodenum



	Major pancreatic duct	Minor pancreatic duct
Horse	Present	Present
Dog	Present	Present
Ox	May abscent	Present
Pig	Abscent	Present
Small ruminant	Present	Abscent

Pancrease

- **Major Pancreatic duct:**

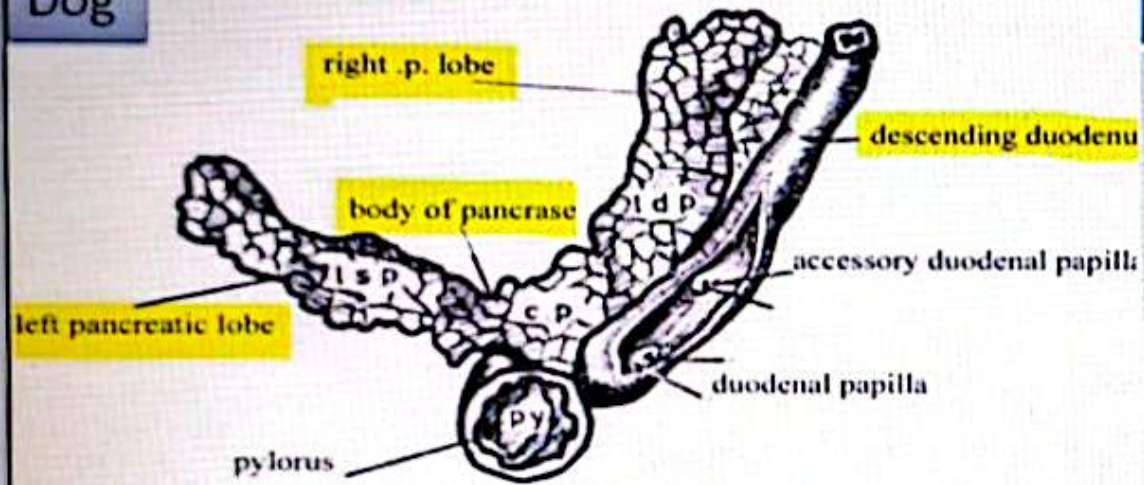
- ✓ Present in: horse + dog + small ruminants
- ✓ ~~Abscent in: pig~~
- ✓ **May** absent in: ox

- **Accessory (minor) pancreatic duct:**

- ✓ Present in: horse + dog + ox ~~+ pig~~
- ✓ Abscent in: small ruminants

Species	Pancreatic Duct	Accessory Pancreatic Duct
Horse	+	+
Ox	Usually absent	+
Pig	-	+
Small ruminants	+	-

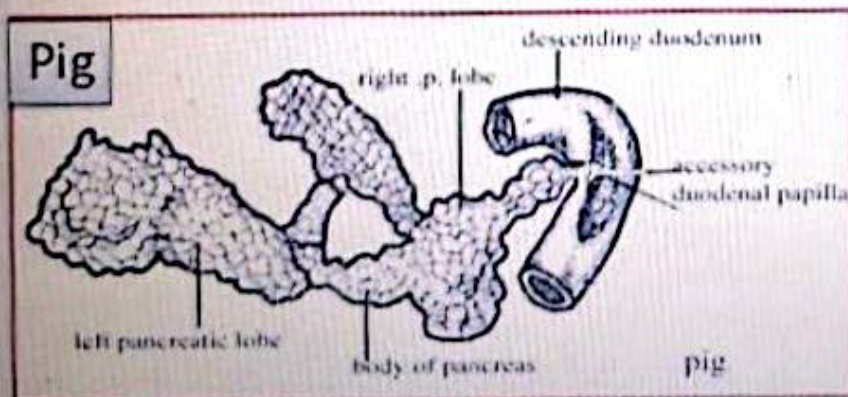
Dog



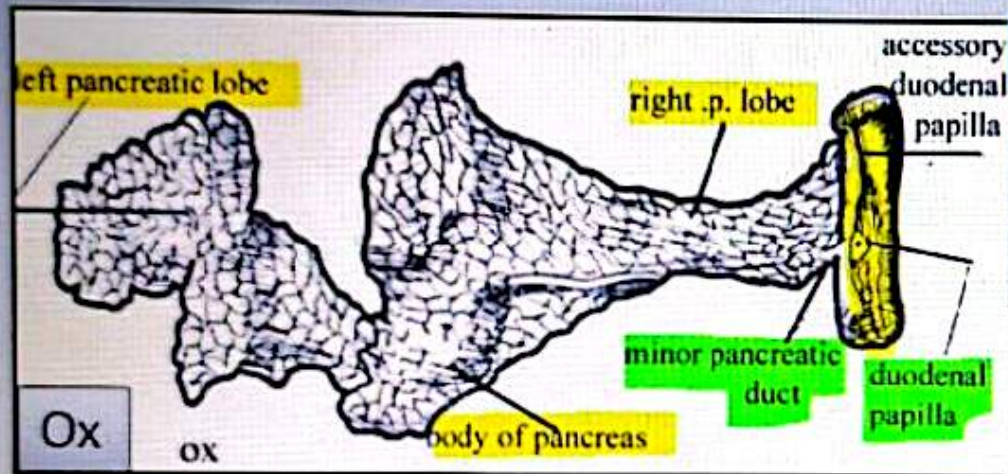
Dog

- Centrally situated
- U-shape
- Body form **pancreatic notch**
- Left: related to L.K.
- Right: extend along descending duodenum
- Major: present
- Minor: present

Pig

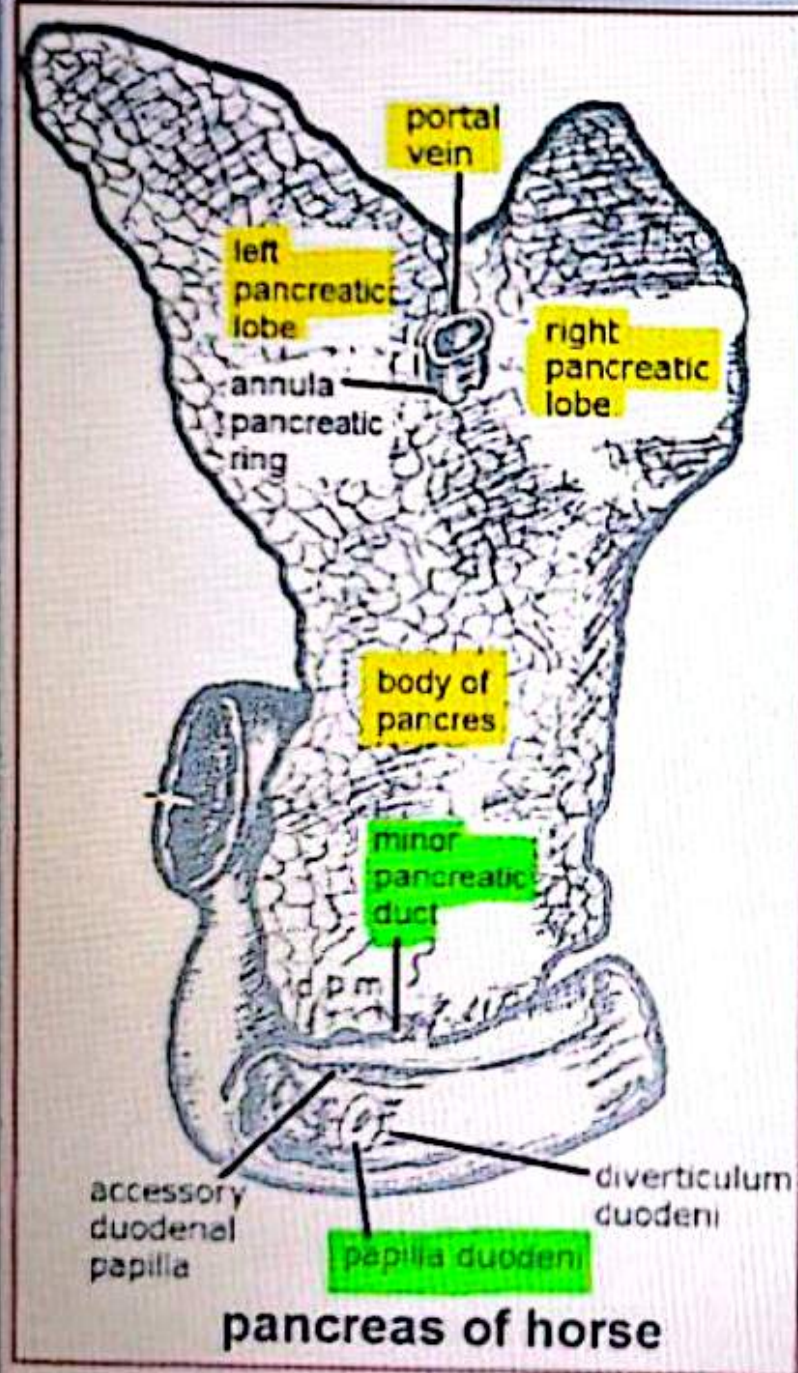


Ask AI



Ox

- Body small
- Form **deep pancreatic notch**
- Left: small & related to left crura of diaphragm & dorsal ruminal sac
- Right: extend along **descending duodenum**
- Major: may absent
- Minor: present



Horse

- Portal ring
- Large body
- Left: large
- Right: short
- Major: present
- Minor: present